



## Maternal $\beta$ -hydroxybutyrate during the dry period is associated with altered epithelial cell regulation and rumen morphology of offspring as calves

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### ABSTRACT

The dry period is a critical stage for fetal development, during which maternal nutritional and metabolic status are strongly associated with postnatal growth and organ development in calves. Among metabolic indicators, elevated maternal serum BHB concentrations during this period have attracted increasing attention; however, their relationships with offspring rumen development remain poorly defined. This study aimed to investigate the effect of high maternal BHB concentrations during the dry period on rumen development in neonatal and 1-mo-old calves. Forty-eight Holstein cows with similar BW, parity, and expected calving date were monitored throughout the dry period. Based on serum BHB concentrations measured at wk -7, -5, -3, and -1 during the dry period, cows were classified into a low-BHB group (CON;  $0.23 \pm 0.02$  mM;  $n = 24$ ) or a high-BHB group (MB;  $0.69 \pm 0.03$  mM;  $n = 24$ ). Rumen tissues were collected from calves at birth and at 1 mo of age to assess morphological development and inflammatory responses. Maternal BHB concentrations did not affect calf birth weight; however, calves from MB cows showed association with increased rumen mass but reduced papilla length and width, indicating association with impaired structural development. Rumen epithelial integrity related markers were altered in the MB group, as evidenced by downregulation of the tight junction protein Claudin1 and upregulation

of inflammatory cytokines. Furthermore, expression of peroxisome proliferator-activated receptor (*PPAR*)  $\gamma$  was markedly reduced in MB calves, consistent with morphological alterations and inflammation. In conclusion, high maternal BHB concentrations during the dry period may impair rumen development in offspring calves potentially by influencing pathways related to epithelial barrier-related markers, altering metabolic gene regulation, and promoting *PPAR* $\gamma$  mediated inflammatory signaling. These findings emphasize the importance of nutritional strategies during the dry period to mitigate maternal metabolic disturbances and support optimal rumen epithelial development and function in offspring calves.

**Key words:** dry period,  $\beta$ -hydroxybutyrate, maternal, calf, rumen

### INTRODUCTION

The nonlactating (dry) period (~8 wk before calving) is critical for both mammary gland regeneration and healthy fetal development in the next production cycle

(Neville, 1974; Sørensen and Enevoldsen, 1991). During this stage of rapid fetal growth and organogenesis, even minor changes in maternal metabolic status can have long-lasting effects on the growth, maturation, and function of fetal tissues (McLaren, 1965; Nosten, 1989; Reynolds et al., 2022). The transition to lactation, encompassing late gestation and early postpartum, imposes substantial metabolic demand and commonly induces negative energy balance (NEB), characterized by increased adipose lipolysis and higher circulating BHB, the predominant ketone body and a recognized biomarker of ketosis in dairy cows (Duffield et al., 2009; Nazeer et al., 2019). Although the effects of elevated BHB on maternal health are well documented (Mann et al., 2017; Sangalli et al., 2025), the downstream consequences for offspring

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growth and organ development remain insufficiently defined.

The rumen, the principal digestive organ in ruminants, is essential for postnatal nutrient utilization and growth (Baldwin and Connor, 2017). At birth, the rumen is under developed and poorly keratinized, which limits VFA absorption (Hinders and Owen, 1965; Ghaffari et al., 2025). The thickness and differentiation of the ruminal epithelium, particularly the stratum corneum, are key indicators of functional maturity and can be enhanced by appropriate nutritional strategies (Ragionieri et al., 2016). Early rumen development is highly sensitive to maternal metabolic influences as well as postnatal diet (Reynolds et al., 2022). Despite extensive research on dietary drivers of rumen maturation, it remains unknown whether maternal BHB levels during the dry period program neonatal rumen structure and epithelial function.

Rumen epithelial proliferation, differentiation, and barrier function are regulated by multiple signaling pathways, including the PPAR, Hippo, Wnt/ $\beta$ -catenin, and NF- $\kappa$ B pathways (Mandard et al., 2004; Connor et al., 2013, 2014). Among these pathways, PPAR signaling, particularly PPAR $\gamma$ , is central to epithelial renewal, lipid metabolism, and barrier integrity (Sun et al., 2021b). Downstream effectors include MYC, CCND1, AXIN, TGF $\beta$ 1, and LEF1/TCF, which promote epithelial turnover (Connor et al., 2013; Steele et al., 2015); SLC16A1 (MCT1), which facilitates short-chain fatty acid absorption (Sprowl-Tanio et al., 2016); CDH1, critical for cell adhesion and barrier stability (Baldwin and Connor, 2017; Aschenbach et al., 2019); and HMGCS2, the rate-limiting enzyme for ruminal ketogenesis (Zhuang et al., 2023). Beyond epithelial metabolism, PPAR $\gamma$  also modulates inflammatory responses by suppressing NF- $\kappa$ B signaling, reducing cytokine production, and modulating mucin secretion (Dreyer et al., 1992; Blanquart et al., 2003; Zhao et al., 2018). Elevated BHB likewise acts as a signaling molecule that influences mucosal immunity and microbial composition, as shown by reduced microbial diversity and increased pathogenic bacteria following intravenous BHB infusion in yaks (Qi et al., 2021). Conversely, elevated maternal BHB has been associated with NF- $\kappa$ B activation, NLRP3 inflammasome stimulation, and caspase-dependent apoptosis (Alharthi et al., 2021; Wang et al., 2025), processes that may impair tissue development and immune competence. However, the association between maternal BHB concentrations during the dry period and the inflammatory tone and basal inflammatory gene expression patterns in the calf rumen remains unclear.

We therefore hypothesized that high maternal BHB during the dry period disrupts key regulatory networks, particularly PPAR $\gamma$  signaling and its interaction with inflammatory pathways, which govern rumen epithelial

development and barrier function, ultimately impairing rumen epithelial barrier integrity and absorptive function. To test this hypothesis, the objective of this study was to evaluate whether high maternal BHB during the dry period is associated with changes in calf rumen morphology, epithelial differentiation, and the expression of key inflammation-related genes and proteins associated with barrier integrity, lipid metabolism, and inflammatory regulation in neonatal and weaned calves born to cows with different maternal BHB concentrations during the dry period.

## MATERIALS AND METHODS

### *Animals and Experimental Design*

The study was conducted at a commercial dairy farm (HSMD Dairy, Jinan, Shandong Province, China) from September to December 2023. All experimental procedures involving animals were approved by the Institutional Animal Care and Use Committee of Nanjing Agricultural University (IACUC protocol #20230625N12) and performed in accordance with the Guidelines for the Care and Use of Experimental Animals. Investigators were blinded to group assignment throughout all experimental procedures and data analysis.

A total of 60 multiparous Holstein dairy cows were enrolled in a companion study at the same commercial dairy farm, where all cows were managed under identical dietary, housing and management conditions. The present study represents a retrospective, secondary analysis of a predefined subset of this cohort. From the original 60 cows, 48 were selected based on the following inclusion criteria: comparable BCS (a low serum BHB group, **CON**:  $3.35 \pm 0.06$ ; a low serum BHB group, **MB**:  $3.39 \pm 0.07$ ;  $P = 0.96$ , Table 1), parity, and expected calving dates; availability of complete serum BHB measurements during the dry period; successful delivery of a single live calf; and absence of clinically diagnosed metabolic or reproductive disorders during the postpartum period (e.g., clinical ketosis, hypocalcemia, displaced abomasum). Clinical ketosis was defined based on veterinary diagnosis and farm health records, using the presence of characteristic clinical signs, and no biochemical screening for subclinical ketosis was performed postpartum. The remaining 12 cows were excluded due to incomplete biochemical data, sample loss ( $n = 3$ ) or quality control failure ( $n = 8$ ), or nonketotic postpartum clinical complications ( $n = 1$ ), rather than experimental outcomes of interest.

Based on previously published literature describing serum BHB as an indicator of prepartum metabolic status in dairy cows (Fürl, 2016; Couperus et al., 2021; Sun et al., 2021a), blood samples were collected from the coc-

**Table 1.** Maternal growth performance during the dry period

| Maternal cow item          | Treatment <sup>1</sup> |       |       | P-value   |        |                  |
|----------------------------|------------------------|-------|-------|-----------|--------|------------------|
|                            | CON                    | MB    | SEM   | Treatment | Time   | Treatment × time |
| Cow BW (kg)                | 742.9                  | 749.1 | 17.16 | 0.72      | 0.46   | 0.77             |
| Cow BCS                    | 3.30                   | 3.33  | 0.13  | 0.85      | 0.65   | 0.96             |
| Cow BW change (kg/wk)      | −19.0                  | −30.1 | 4.5   | 0.38      | 0.14   | 0.49             |
| Cow BCS change (points/wk) | −0.03                  | −0.04 | 0.01  | 0.20      | 0.02   | 0.39             |
| Serum BHB (mM)             | 0.23                   | 0.69  | 0.03  | <0.0001   | <0.001 | <0.001           |

<sup>1</sup>CON = calves born to cows with low serum BHB concentrations; MB = calves born to cows with high serum BHB concentrations.

cygeal vein at wk −7, −5, −3, and −1 of the dry period using 10-mL vacuum tubes with clot activator. Serum BHB concentrations were measured using validated test strips (Abbott Diabetes Care Inc., Alameda, CA) for preliminary group validation and a commercially available bovine sandwich ELISA kit (catalog no. #MM-5100301, Meimian, China), according to the manufacturer's instructions. The intra-assay and interassay CV were 2.5% and 1.5%, respectively. Based on average serum BHB concentrations, enrolled cows were assigned to CON or MB using the median (50th percentile) serum average BHB concentration of each cow in the study population (0.45 mM) as the cutoff. Accordingly, cows with serum BHB ≤0.45 mM (0.23 ± 0.02 mM; n = 24) were classified as CON, whereas cows with serum BHB >0.45 mM (0.69 ± 0.03 mM; n = 24) were classified as MB. The prepartum BHB concentrations observed in this study were below thresholds typically associated with clinical ketosis. No clinical disease events were recorded during the experimental period.

For rumen analysis, only dam-calf pairs with quality control-verified rumen tissue samples were included to ensure consistent sampling location, RNA integrity, and histological preservation. These samples were balanced by maternal BHB group and sampling age (birth and 1 mo; n = 12 per stratum). These 48 dams were in the dry period, with similar BW (mean ± SE; 746.0 ± 8.50 kg) and parity (1.95 ± 0.16), and all carried single male fetuses, as confirmed by B-mode ultrasonography. Within each group, cows were randomly allocated to neonatal (n = 12) or 1-mo-old (n = 12) sampling groups (Figure 1A).

All cows were housed in freestall barns bedded with commercial bedding material, had ad libitum access to clean water, and received a TMR twice daily (0700 and 1700 h), formulated according to National Research Council recommendations (NRC, 2001) and Du et al. (2018). The chemical composition was reported in Supplemental Table S1 (see Notes). The analysis was performed as before (Gai et al., 2025). Body weight and BCS (1–5 scale, 1 = thin and 5 = obese; scored by 2 independent evaluators) were recorded weekly (Edmundson et al., 1989). Calves received colostrum (100 g/kg

BW) immediately after birth, followed by 2 additional feedings within the first 12 h. Standard calf management protocols were followed thereafter.

### Sample Collection

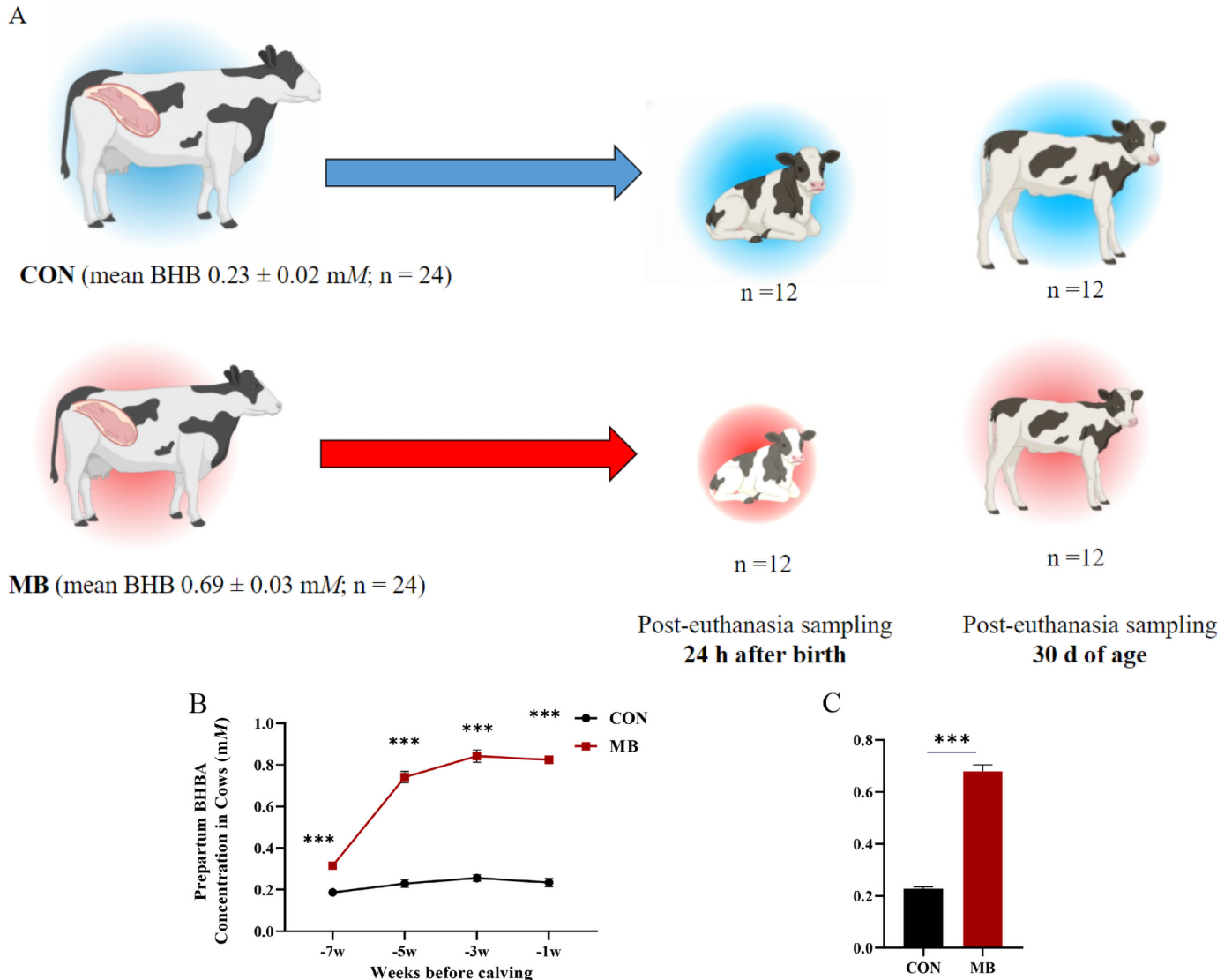
Calves were killed at either 24 h after birth or at 1 mo of age by intravenous injection of 10 mL of pentobarbital sodium. Rumen tissues were excised, weighed, and tissues from the ventral sac of the rumen were either fixed in 4% paraformaldehyde (PFA, G1101; Servicebio) for histological analysis or snap-frozen in liquid nitrogen for subsequent assays.

### Histological Sample Preparation

Fresh ruminal tissue samples were immediately rinsed 3 times with ice-cold PBS (pH 7.2–7.4) to remove residual digesta from the mucosal surface. The tissues were then trimmed into standardized blocks (1 cm × 1 cm × 0.3 cm; length × width × thickness) using sterile disposable scalpels. The tissue blocks were fixed in 4% PFA at room temperature (22°C–25°C) for 6 to 8 h. After fixation, the tissues were wrapped in autoclaved gauze and placed in perforated glass bottles, then rinsed under a slow stream of prechilled (4°C) running water for 2 to 3 h. Subsequently, the tissues underwent graded ethanol dehydration (50%, 70%, 80%, 90%, 95%, and 100%; 1 h each), were cleared in xylene (2 × 30 min), and embedded in paraffin wax at 60°C ± 1°C with infiltration for 12 h. Serial sections of 4- to 5-μm thickness were cut using a rotary microtome (Leica RM2255, Wetzlar, Germany). Sections were floated on a 45°C water bath, mounted onto poly-L-lysine-coated glass slides (Sigma-Aldrich, St. Louis, MO), and dried overnight at 37°C in a slide-drying oven.

### Hematoxylin and Eosin Staining and Morphometric Analysis

Deparaffinized sections were rehydrated through a descending ethanol series. Hematoxylin staining was



**Figure 1.** (A) Experimental design and sampling. Dams were classified by the mean serum BHB concentration across the dry period (samples at  $-7$ ,  $-5$ ,  $-3$ , and  $-1$  wk prepartum) into CON ( $\leq 0.45$  mM;  $n = 24$ ) or MB ( $> 0.45$  mM;  $n = 24$ ). Male calves from each group were randomized to terminal sampling at 24 h after birth or at 30 d of age ( $n = 12$  per group per age; distinct calves). Created in Biorender.com. Zheng, J. (2026). (B) Temporal changes in maternal serum BHB concentrations during the dry period in dairy cows. Average serum BHB concentrations throughout the entire dry period. (C) Serum BHB concentrations in dairy cows of the maternal low-BHB group and maternal high-BHB group during the dry period. CON = calves born to cows with low serum BHB concentrations ( $0.23 \pm 0.02$  mM); MB = calves born to cows with high serum BHB concentrations ( $0.69 \pm 0.03$  mM). Data are means  $\pm$  SEM; ns = not significant. \*\*\* $P < 0.001$ .

performed for 1 min, followed by a 2 min rinse under running tap water. Sections were then counterstained with eosin for 1 min, dehydrated through ascending ethanol concentrations, cleared in xylene, and cover slipped with neutral balsam mounting medium (Sigma-Aldrich). Histological morphology was examined under a light microscope (e.g., Olympus BX53, Tokyo, Japan) using a  $4\times$  objective, and higher magnification ( $10\text{--}40\times$ ) was used to confirm epithelial layer identification. Image analysis was performed using ImageJ software (National Insti-

tutes of Health, Bethesda, MD; version 1.54g). A  $100\text{-}\mu\text{m}$  scale bar was used to calibrate image dimensions. Morphometric parameters, including papilla height, width, and epithelial thickness, were measured using consistent thresholds and analytical modules across all samples. Papillae distribution was visually assessed from hematoxylin and eosin-stained sections to provide qualitative structural context. For each morphometric parameter, 3 nonoverlapping fields were analyzed per sample, and 5 replicate measurements were taken per field to account



for intrasample variability (Feldman and Wolfe, 2014). Rumen epithelial layers were differentiated according to established histological definitions (Goosen, 1976; Anison and Bryden, 1998; Graham and Simmons, 2005; Ragionieri et al., 2016; Baldwin and Connor, 2017). The stratum basale was identified as the innermost layer composed of a single row of cuboidal to columnar cells adjacent to the basement membrane with densely packed nuclei. The stratum spinosum consisted of several layers of polyhedral cells exhibiting prominent intercellular bridges. The stratum granulosum comprised 1 to 3 layers of flattened cells containing intensely eosinophilic keratohyalin granules. The stratum corneum formed the outermost layer and was composed of flattened, fully keratinized, anucleated cells with a pale eosinophilic appearance.

### Real-Time Quantitative PCR Analysis

Total RNA was isolated from rumen tissue samples treated with DNase following the standard protocol to remove any contaminating genomic DNA and using TRIzol reagent (Invitrogen, Grand Island, NY) according to the manufacturer's instructions. The concentration and purity of the extracted RNA were determined using a NanoDrop ND-1000 spectrophotometer (Thermo Fisher Scientific, Waltham, MA), with acceptable A260/A280 ratios ranging from 1.9 to 2.0. RNA integrity was further verified by 1% agarose gel electrophoresis, and only samples with clear, intact 28S and 18S rRNA bands were used for subsequent analysis. One microgram (1 µg) of high-quality total RNA from each sample was reverse transcribed into cDNA using the iScript cDNA Synthesis Kit (catalog no. TSK302S; Tsingke Biotechnology Co., Ltd., Beijing, China), following the manufacturer's protocol. Negative control reactions were included in all reverse transcription experiments, in which no reverse transcriptase was added, to ensure that amplification of genes of interest did not occur due to residual genomic DNA in the RNA samples. The thermal cycling conditions for reverse transcription were as follows: 60°C for 1 min (priming), 4°C for 2 min (cooling), 50°C for 15 min (reverse transcription), and final enzyme inactivation at 85°C for 5 s. Bovine-specific primer pairs were designed using the National Center for Biotechnology Information Primer-BLAST tool and synthesized by Tsingke Biotech (Beijing, China). The primer information for the genes is shown in Supplemental Table S2 (see Notes). All primers had an optimal annealing temperature of 60°C. For each quantitative polymerase chain reaction (qPCR), a standard curve ( $R^2 > 0.99$ ) containing all sample range was generated using 5 serial dilutions (5×) of cDNA to test amplification efficiency in accordance with "The MIQE guidelines: minimum information for publication of

quantitative real-time PCR experiments" (Bustin et al., 2009). Only primer sets with a PCR efficiency between 90% and 110% and an  $R^2 > 0.985$  were used for subsequent qPCR experiments. All primer specificities and performance validations were conducted on an Mx3000P qPCR instrument (Agilent Technologies) using cDNA synthesized from rumen tissue samples stored in the tissue repository. Primer pairs for both target genes and reference genes were analyzed on a single reaction plate (typically a 96-well plate). Samples were run in triplicate to ensure the reliability and reproducibility of the results. A melt curve analysis was performed after each qPCR run to confirm amplification of a single target amplicon and to check the specificity of the primers. The qPCR reactions were performed in a total volume of 20 µL using iQ Synergetic Binding Reagent Green Supermix (Tsingke Biotechnology Co. Ltd., catalog no. TSE201) on a Bio-Rad CFX96 qPCR System. The thermal cycling program was as follows: initial denaturation at 95°C for 5 min, followed by 40 amplification cycles, each consisting of denaturation at 95°C for 10 s and annealing and extension at 60°C for 30 s (Quant Studio 5–1304553S).

The mRNA expression levels were quantified using iQTM SYBR Green Supermix (#TSE201; Tsingke, China) on an Applied Biosystems 7500 qPCR System, and data were analyzed with Bio-Rad CFX Manager 3.1 software. Gene expression was calculated using the  $2^{-\Delta\Delta Ct}$  method (Ontsouka et al., 2004). Both *18S rRNA* and *GAPDH* (glyceraldehyde-3-phosphate dehydrogenase) were used as validated reference genes, as described in our previous studies and others. These reference genes have been validated for stability in bovine rumen tissue under similar physiological conditions (Huggett et al., 2005; Gai et al., 2025). In the present study, both reference genes were used for normalization to increase the accuracy and reliability of gene expression quantification.

### Western Blot Analysis

Total protein was isolated from 100 mg of ground rumen tissue using 1 mL of ice-cold radioimmunoprecipitation assay lysis buffer (catalog no. BL504A; Biosharp) supplemented with 1 mM protease inhibitor phenylmethylsulfonyl fluoride (catalog no. L507A; Biosharp). The homogenates were incubated on ice for 20 min and centrifuged at  $12,000 \times g$  for 20 min at 4°C. The protein loading volume was 15 µL per well. Proteins were separated by SDS polyacrylamide gel electrophoresis (catalog no. 89888; Thermo Fisher) and transferred onto polyvinylidene fluoride membranes (catalog no. PVH10100; Millipore). Membranes were blocked with 1.5% skim milk for 2 h at room temperature, then incubated overnight at 4°C with primary antibodies. Based on their established roles in rumen epithelial stress response, barrier integrity, and

lipid metabolism, the following antibodies were used: anti-Mucin 1 (1:800, #A0333, ABclonal, China), rabbit anti-PPAR $\gamma$  (1:1,000, #16643-1-AP, Proteintech, China), rabbit anti-Claudin 1 (1:1,000, #28674-1-AP, Proteintech, China), and  $\beta$ -tubulin (1:1,000, AF2835, Beyotime, China). After incubation with horseradish peroxidase (HRP)-conjugated secondary antibodies, either HRP-conjugated goat anti-rabbit IgG (H<sup>+</sup>L; 1:1,500, A0208, Beyotime) or HRP-conjugated goat anti-mouse IgG (H<sup>+</sup>L) (1:1,500, A0216, Beyotime, China) for 2 h at room temperature, protein bands were detected using enhanced chemiluminescence (RM00021P, ABclonal, China) and imaged on a ChemiDoc MP system (Bio-Rad). Band intensities were quantified using Protein Simple Systems (Protein Simple, Santa Clara, CA) and normalized to  $\beta$ -tubulin as an internal control. After washing with Tris-buffered saline containing 0.1% Tween 20 (1 $\times$  TBST) 6 times (10 min per wash), membranes were incubated with HRP-labeled secondary antibodies (HRP-labeled goat anti-rabbit antibody and HRP-labeled goat anti-mouse antibody) for 2 h at room temperature. The membranes were then washed with 1 $\times$  TBST 6 times (10 min per wash). Bands were visualized using the enhanced chemiluminescence Plus Kit (Vazyme, Nanjing, China), and signals were obtained using the ChemiDoc MP System (Bio-Rad, Berkeley, CA). Band intensities were analyzed using Image Lab software (Bio-Rad, Berkeley, CA) and normalized to  $\beta$ -tubulin. Western blotting provided quantitative validation of protein expression.

### Statistical Analysis

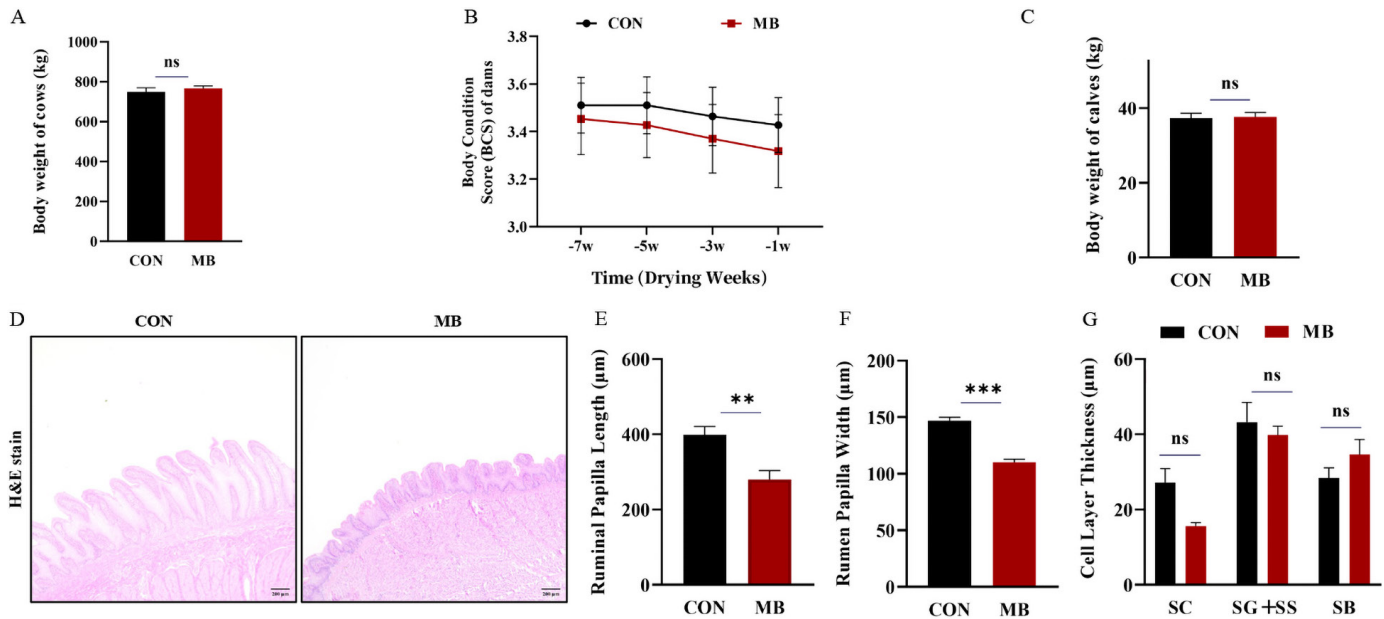
Statistical analyses were performed in R v4.3.1 (R Core Team, 2023) using *lme4* v1.1-35 and *lmerTest* v3.1-3. Figures were produced in GraphPad Prism 8.0 (GraphPad Software, San Diego, CA). Before the experiment, a power analysis was conducted using the *simr* package in R to determine the minimum sample size required to detect a biologically relevant difference in maternal serum BHB, which was used to define maternal metabolic status and treatment grouping. Based on prior on-farm variability and physiological ranges in maternal serum BHB, the analysis indicated that a sample size of  $n > 5$  cows per group would provide ~80% power at an  $\alpha$  level of 0.05 using the model described below. The power analysis was therefore intended to justify the adequacy of the sample size for detecting separation in maternal metabolic status, rather than to individually power each downstream calf outcome. Maternal repeated-measures outcomes (serum BHB, BW, BCS) were analyzed using linear mixed-effects models with fixed effects of treatment (maternal BHB level), week, and their interaction, and cow included as a random effect. Model fit was

evaluated using Akaike and Bayesian information criteria. Calf outcomes measured once per animal (e.g., birth BW, rumen mass, histomorphometry, western blot, and gene expression) were analyzed using linear fixed-effects models including maternal BHB group as a fixed effect. These calf rumen outcomes were considered secondary and exploratory endpoints within a developmental programming framework and were not individually powered, due to the absence of reliable prior variance estimates for rumen structural and molecular markers. Quantitative PCR data were analyzed on  $\Delta$ cycle threshold values (or log<sub>2</sub>-transformed expression where indicated), and protein densitometry data were log-transformed when residuals were skewed. Model assumptions were evaluated using Shapiro–Wilk tests for normality and Levene’s tests for homogeneity of variances on model residuals. When assumptions were violated, Box–Cox-guided transformations were applied. For repeated-measures data, MB versus CON contrasts were evaluated within each week when the time  $\times$  treatment interaction was significant, and *P*-values were adjusted using the Holm method. Denominator degrees of freedom and *P*-values were obtained by Satterthwaite approximation (lmerTest). Results are presented as means  $\pm$  SEM; significance was set at  $P < 0.05$ , and trends were defined as  $0.05 \leq P < 0.10$ .

## RESULTS

### Maternal BHB Levels, Neonatal Calf Growth, and Rumen Development

During the dry period, maternal serum BHB concentrations averaged 0.23 mM in the CON group and 0.69 mM in the MB group, with the overall BHB concentration range throughout the study spanning from 0.11 to 1.01 mM ( $P < 0.01$ ; Table 1, Figure 1B, C, Supplemental Table S3, see Notes). In the CON group, BHB concentrations ranged from 0.11 to 0.53 mM, with mean values of 0.19, 0.23, 0.26, 0.24 mM at 7, 5, 3, 1 wk before calving, respectively. In contrast, the MB group exhibited higher BHB concentrations, ranging from 0.20 to 1.01 mM, with corresponding values of 0.32, 0.74, 0.86, and 0.82 mM at the same time points. As reported previously (Duffield, 2000; Duffield et al., 2009; Nazeer et al., 2019), these levels were below the clinical threshold for ketosis. Maternal BW, BCS, BW change, and BCS change did not differ between those 2 groups ( $P > 0.1$ , Table 1, Figure 2A, B). Calf birth weights were similar between CON and MB groups ( $P = 0.85$ ; Supplemental Table S4, see Notes; Figure 2C). However, rumen mass was significantly greater in MB calves ( $P = 0.03$  and  $P = 0.01$ ; Supplemental Table S4).



**Figure 2.** Observed growth performance and rumen morphological characteristics of neonatal calves born to dams with low (CON) or high (MB) serum BHB concentrations during the dry period. (A) Body weight of cows in the neonatal group. (B) Changes in BCS during the dry period. (C) Neonatal BW at birth. (D) Representative hematoxylin and eosin (H&E) staining of ruminal epithelium (scale bar = 200  $\mu$ m). (E) Rumen papilla length and (F) width. (G) Thickness of individual rumen epithelial layers, including stratum corneum (SC), stratum granulosum + stratum spinosum (SG + SS), and stratum basale (SB). Data are means  $\pm$  SEM ( $n = 12$  calves per group). CON = calves born to cows with low serum BHB concentration; MB = calves born to cows with high serum BHB concentration. Data are means  $\pm$  SEM; ns = not significant. \*\* $P < 0.01$ , \*\*\* $P < 0.001$ .

### Structure, Gene Expression, and Protein Expression Are Altered in CON Versus MB Newborn Calves

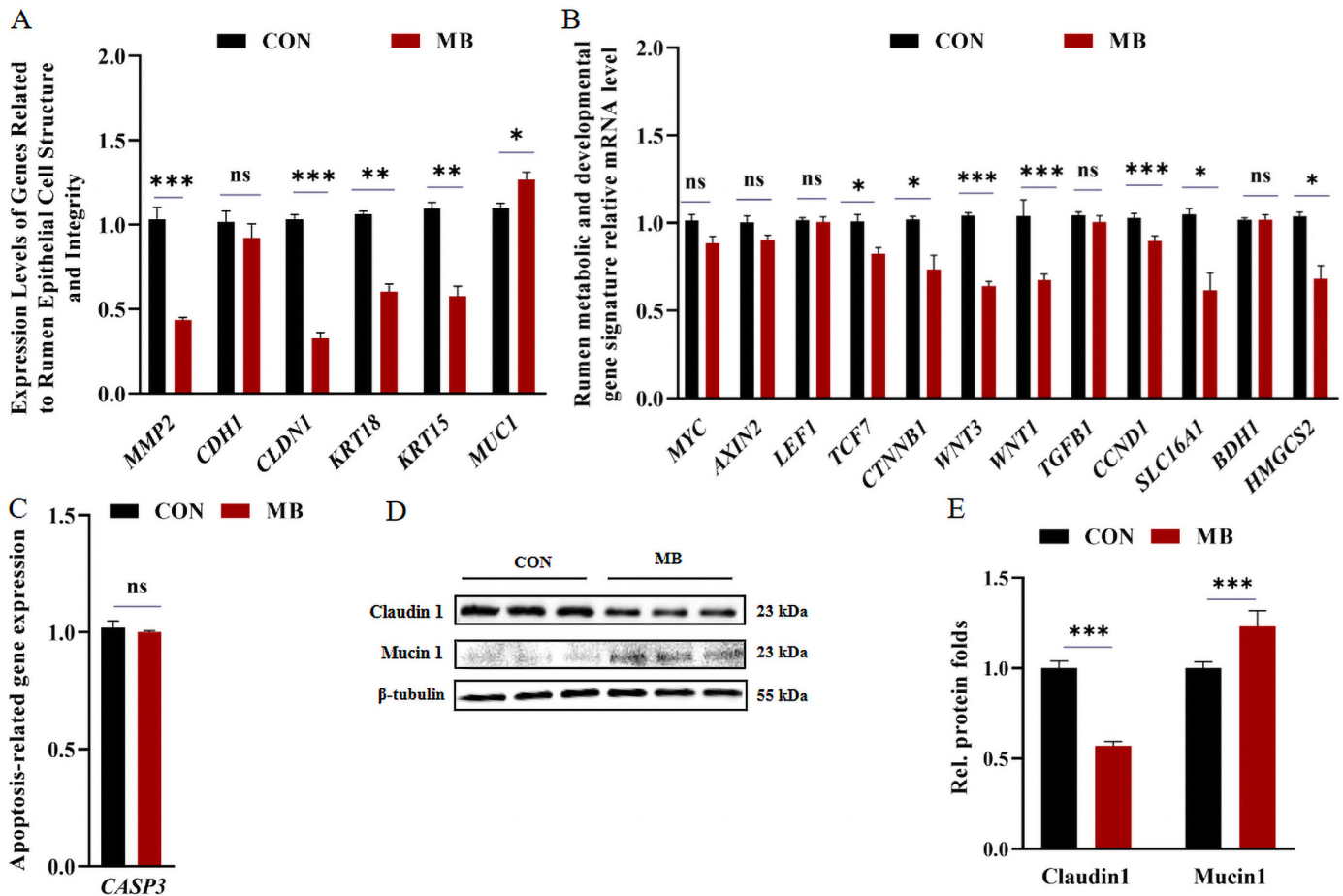
At birth, calves born to MB dams had significantly shorter fixed rumen papillae and narrower papilla width compared with CON calves ( $P < 0.05$ ; Figure 2D–F; Supplemental Table S4). However, there were no significant differences in the thickness of rumen epithelial cell layers, including the stratum granulosum, stratum spinosum, and stratum basale, between the CON and MB groups ( $P \geq 0.22$ , Figure 2G; Supplemental Table S4). The MB group was associated with a tendency toward reduced stratum corneum thickness in neonatal calves ( $P = 0.07$ ; Figure 2G; Supplemental Table S4), showing a numerically large shift of  $\sim 36\%$  decrease compared with CON calves.

Rumen epithelial integrity is essential for maintaining barrier function against digestive and environmental stressors. In MB calves, expression of key rumen epithelial barrier integrity related genes, including *MMP2*, *CLDN1*, *KRT18*, and *KRT15*, was significantly reduced ( $P < 0.05$ ; Figure 3A). Similarly, rumen keratin-associated genes, including *CLDN1*, *KRT18*, and *KRT15*, were also downregulated in the MB group relative to the CON group ( $P < 0.05$ ; Figure 3A). In contrast, expression of *MUC1*, which encodes a transmembrane mucin involved in epithelial protection, was significantly increased in

MB calves ( $P < 0.05$ ; Figure 3A). Consistent with this, MB calves exhibited decreased expression of genes associated with rumen epithelial cell proliferation and differentiation, including *TCF7*, *CTNNB1*, *WNT3*, *WNT1*, *TGFB1*, and *CCND1* ( $P < 0.05$ ; Figure 3B), as well as reduced expression of ketone body metabolism related genes (*SLC16A1* and *HMGCS2*) ( $P < 0.05$ ; Figure 3B). No significant difference was detected in the apoptosis marker *CASP3* ( $P > 0.05$ ; Figure 3C). Protein-level analyses using western blot confirmed that Claudin1 expression was decreased, whereas MUC1 expression was increased in MB calves ( $P < 0.05$ ; Figure 3D–E), consistent with mRNA expression patterns.

### Structure, Gene Expression, and Protein Expression Are Altered in CON Versus MB 1-Mo-Old Calves

Similarly, maternal BW and BCS did not differ between the 2 groups during the dry period ( $P > 0.1$ , Table 1, Figure 4A, B). At 1 mo of age, calf BW remained unaffected between offspring of dams with differing BHB levels ( $P = 0.55$ , Figure 4C, Supplemental Table S4). However, based on measurements from fixed tissues, rumen papilla length ( $P = 0.01$ ) and width ( $P = 0.05$ ) remained reduced in calves born to MB dams relative to CON dams (Figure 4D–F, Supplemental Table S4). No significant differences were observed in the thickness



**Figure 3.** Rumen epithelial integrity and barrier function indicators in neonatal calves born to dams with low (CON) or high (MB) serum  $\beta$ -hydroxybutyrate (BHB) concentrations during the dry period. (A) mRNA expression of genes associated with epithelial barrier integrity. (B) mRNA expression of genes involved in rumen epithelial differentiation and ketone body metabolism. (C) mRNA expression of the apoptosis-related gene *CASP3*. (D) Western blot detection of Claudin1 and Mucin1 proteins normalized to  $\beta$ -tubulin. (E) Relative (Rel.) protein abundance of Claudin1 and Mucin1. CON = calves born to cows with low serum BHB concentration; MB = calves born to cows with high serum BHB concentration. Relative expression values were normalized to the CON group. CON group was set to 1. MB was expressed relative to CON. Data are means  $\pm$  SEM; ns = not significant. \* $P < 0.05$ , \*\* $P < 0.01$ , \*\*\* $P < 0.001$ .

of the rumen epithelium, including the granular layer (stratum granulosum, **SG**) and spinous layer ( $P = 0.36$ ), as well as the basal layer (stratum basale, **SB**;  $P = 0.89$ ; Supplemental Table S4, Figure 4D and G), consistent with previous studies on neonates. A similar trend was observed in 1-mo-old calves as in neonatal calves, in which MB was associated with an  $\sim 23\%$  reduction in stratum corneum thickness compared with CON calves ( $P = 0.052$ ; Figure 4G; Supplemental Table S4).

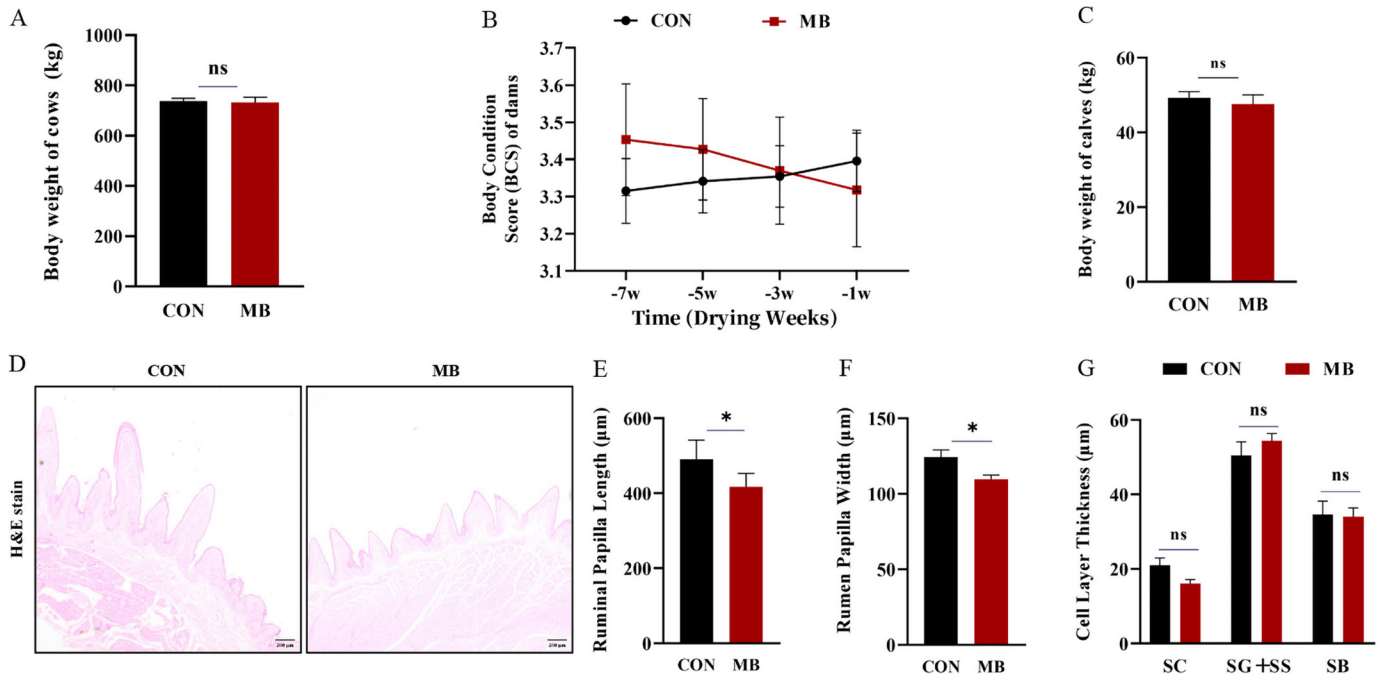
Gene expression analyses revealed that MB calves continued to display reduced expression of epithelial integrity related genes, including *MMP2*, *CDH1*, *CLDN1*, *KRT18*, *KRT15*, and *MUC1* ( $P < 0.05$ ; Figure 5A), rumen epithelial cell proliferation and differentiation related genes, including *MYC*, *AXIN2*, *LEF1*, *TCF7*, *CTNNB1*, *WNT3*, *WNT1*, *CCND1* ( $P < 0.05$ ; Figure 5B), and ketogenesis-related genes, including *SLC16A1* and

*HMGS2* ( $P < 0.05$ ; Figure 5B). Apoptosis-related gene *CASP3* expression remained unchanged (Figure 5C). At the protein level, MB also decreased Claudin1 ( $P < 0.05$ ; Figure 5D–E), while increased Mucin1 ( $P < 0.05$ , Figure 5D–E), mirroring neonatal results.

### Rumen Inflammation Is Altered in Calves Born to Dams with High BHB

To investigate whether varying maternal BHB levels during the dry period indirectly influence the expression of inflammatory factors in offspring rumen, this study evaluated the expression of key inflammation-related genes. In neonatal calves, the expression of *IL1B*, *NFKB1*, *CCL2*, *IL6*, and *NLRP3* was significantly higher in the MB group compared with the CON group ( $P < 0.05$ ; Figure 6A). At 1 mo of age, the levels of *TNF*,





**Figure 4.** Body weight and rumen morphological characteristics of 1-mo-old calves born to dams with low (CON) or high (MB) serum BHB concentrations during the dry period. (A) Body weight of cows in 1-mo-old group. (B) Changes in BCS during the dry period. (C) Calf body weight at 1 mo of age. (D) Representative hematoxylin and eosin (H&E) staining of ruminal epithelium (scale bar = 200  $\mu$ m). (E) Rumen papilla length and (F) width. (G) Thickness of rumen epithelial cell layers, including the stratum corneum (SC), stratum granulosum + stratum spinosum (SG + SS), and stratum basale (SB). CON = calves born to cows with low serum BHB concentrations; MB = calves born to cows with high serum BHB. Data are means  $\pm$  SEM; ns = not significant. \* $P < 0.05$ .

*NFKB1*, *CCL2*, and *IL6* remained elevated ( $P < 0.05$ ; Figure 6B). The persistent upregulation of *MUC1* in MB calves was consistent with the increased inflammatory gene expression.

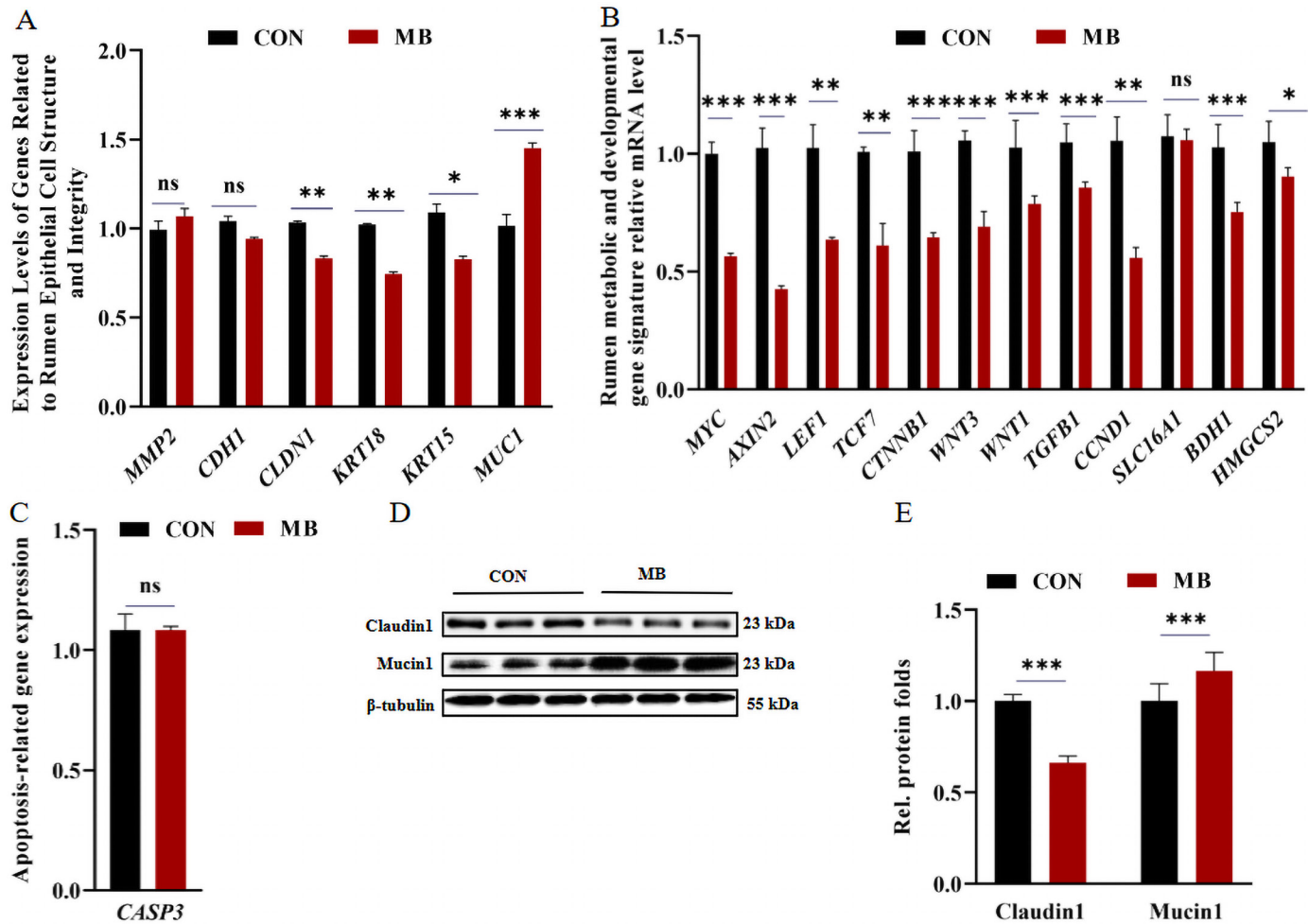
#### Maternal High BHB Is Linked to Alterations in PPAR $\gamma$ (PPARG)-Regulated Inflammatory Signaling in Offspring Calves

The MB calves exhibited reduced expression of fatty acid metabolism related genes relative to CON calves ( $P < 0.05$ ; Figure 6C, D). Among those markers, *PPARG* plays a central role in regulating lipid metabolism, inflammation, and rumen function during rumen development. Protein-level analyses used western blot to determine the expression of *PPARG*. *PPARG* expression was significantly decreased in MB calves at both neonatal and 1-mo-old stages ( $P < 0.05$ ; Figure 6E–G).

### DISCUSSION

During pregnancy, the fetus relies entirely on maternal nutrient supply; therefore, any maternal metabolic disturbance may directly impair fetal development and subsequently affect postnatal growth and performance (Chavatte-Palmer et al., 2018). In cattle, metabolic dis-

turbances such as excessive lipolysis (Gross et al., 2011; Esposito et al., 2014) have been linked to impaired fetal development and reduced postnatal growth rates (Ling et al., 2018).  $\beta$ -Hydroxybutyrate is widely recognized as a sensitive indicator of energy status in cattle, and elevated plasma BHB concentrations have been associated with increased susceptibility to postpartum uterine diseases, particularly endometritis (Bicalho et al., 2017). However, the effect of variation in maternal BHB within the sub-clinical range during the dry period on offspring rumen development remains unclear. In the present study, among dairy cows with plasma BHB concentrations below 1.2 mM, high maternal BHB levels exerted no significant effect on the BW of offspring calves. This finding aligns with previous reports indicating that growth impairments are more apparent under severe NEB or clinical ketosis (Ling et al., 2018). However, notable alterations were observed in the offspring rumen. Specifically, maternal BHB was associated with greater rumen mass likely reflecting increased muscularis and tissue hydration, but retarded the development of rumen papillary length, resulting in shorter, narrower rumen papillae. This was accompanied by a reduction in the expression of tight junction-related genes, which provides indirect evidence of compromised epithelial integrity, and upregulated expression of rumen inflammatory markers. These find-



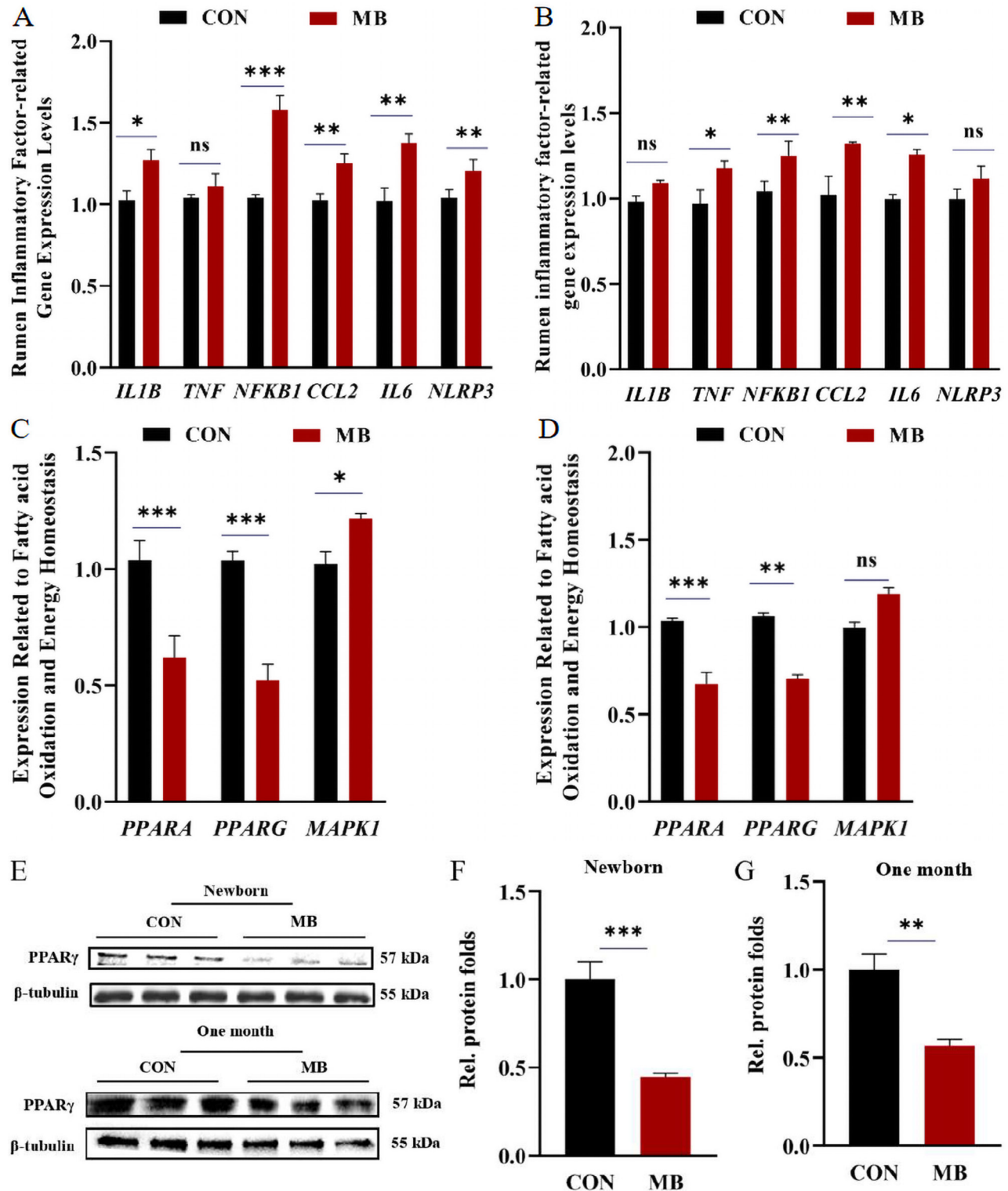
**Figure 5.** Rumen epithelial integrity and barrier function indicators in 1-mo-old calves born to dams with low (CON) or high (MB) serum BHB concentrations during the dry period. (A) mRNA expression of genes related to rumen epithelial integrity. (B) mRNA expression of genes involved in rumen metabolism and epithelial differentiation. (C) mRNA expression of the apoptosis-related gene *CASP3*. (D) Western blot detection of Claudin1 and Mucin1 proteins normalized to  $\beta$ -tubulin. (E) Relative (Rel.) protein abundance of Claudin1 and Mucin1. CON = calves born to cows with low serum BHB concentrations; MB = calves born to cows with high serum BHB. Relative expression values were normalized to the CON group. CON group was set to 1. MB was expressed relative to CON. Data are means  $\pm$  SEM; ns = not significant. \* $P < 0.05$ , \*\* $P < 0.01$ , \*\*\* $P < 0.001$ .

ings demonstrate that even modest increases in maternal BHB below the subclinical range (1.2 mM) during the dry period may still exert critical developmental effects on the calf rumen.

Maternal NEB is a primary driver of high BHB during the dry period (Soares et al., 2021). Previous studies reported that lambs born to NEB-affected ewes had reduced birth and weaning weights (Wang et al., 2024c). In contrast, the absence of BW changes in the current study likely reflects the milder degree of maternal metabolic stress. Although cows in both groups exhibited numerical declines in BW and BCS during the dry period, the magnitude of these changes was modest, did not differ between CON and MB cows, and is consistent with normal maternal tissue mobilization during late gestation rather than treatment-specific effects. Consistent with

the relatively low maternal BHB levels in the current study, calf BW was unaffected; however, rumen mass was significantly increased, suggesting that the rumen is more sensitive to maternal BHB fluctuations than overall body growth. Our results highlight that maternal metabolic stress from high BHB during the dry period, even below clinical thresholds, may program organ-specific developmental outcomes.

The rumen plays a central role in nutrient degradation and absorption in ruminants (Rémond et al., 1995), and its absorptive efficiency depends on well-developed rumen papillae and an intact epithelial structure (Sutton et al., 1963; Dijkstra et al., 1992). Our results support the hypothesis that high maternal BHB during the dry period retarded the development of calves' rumen papillae, potentially reducing absorptive capacity. Comparable



**Figure 6.** PPAR $\gamma$ -related inflammatory responses and metabolic regulatory parameters in offspring calves born to dams with low (CON) or high (MB) serum concentrations during the dry period. (A, B) mRNA expression of inflammation-related genes in the rumen of newborn (A) and 1-month-old (B) calves. (C, D) mRNA expression of genes associated with fatty acid oxidation and energy metabolism in newborn (C) and 1-month-old (D) calves. (E) Western blot detection of PPAR $\gamma$  protein expression in the rumen of newborn and one-month-old calves, normalized to  $\beta$ -tubulin. (F, G) Relative (Rel.) protein abundance of PPAR $\gamma$  in the rumen of newborn (F) and 1-month-old (G) calves. CON = calves born to cows with low serum BHB concentrations; MB = calves born to cows with high serum BHB. Relative expression values were normalized to the CON group. CON group was set to 1. MB was expressed relative to CON. Data are means  $\pm$  SEM; ns = not significant. \* $P$  < 0.05, \*\* $P$  < 0.01, \*\*\* $P$  < 0.001.

reductions in papillae length and epithelial integrity of lambs have been reported in offspring exposed to maternal NEB (Wang et al., 2024c), supporting the consistency of our findings across species. The rumen epithelium consists, which serves as the principal absorptive interface for VFA and other metabolites, is composed of stratified layers, including the stratum corneum, stratum granulosum, stratum spinosum, and stratum basale (Bannink and Tamminga, 2005; Graham and Simmons, 2005). The stratum corneum, the outermost layer, consists of flattened, anucleated, fully keratinized cells that appear pale pink and protect the epithelium from the harsh ruminal environment (Steven and Marshall, 1970; Baldwin, 1998). Beneath this layer, the stratum spinosum and stratum granulosum form intermediate cell layers without distinct boundary (Graham and Simmons, 2005). The stratum spinosum is composed of multiple layers of polygonal cells extending outward from the stratum basale and is characterized by visible clearly defined intercellular bridges under high magnification (Graham and Simmons, 2005). The stratum granulosum typically consists of 1 to 3 layers of flattened cells containing strongly eosinophilic keratohyalin granules in their cytoplasm (Steven et al., 1971; Annison and Bryden, 1998; Baldwin and Connor, 2017). Tight junctions in the stratum spinosum and stratum granulosum contribute to maintaining epithelial barrier integrity. We observed a trend toward reduced stratum corneum thickness and significant downregulation of tight junction proteins, such as Claudin1, as well as keratin-related genes (*KRT18*, *KRT15*), in calves exposed to MB during the dry period. These changes likely impair epithelial integrity and increase rumen permeability, consistent with increased expression of *MUC1*, a mucin associated with epithelial stress responses and mucosal defense (Lavker et al., 1969; Hoorens et al., 2011; Kato et al., 2017). Disruption of similar barrier-related genes has been reported under NEB conditions in lambs, further supporting the biological relevance of these molecular responses (Soares et al., 2021; Wang et al., 2024c). The stratum basale is metabolically active, contributing to nutrient absorption (Annison and Bryden, 1998; Baldwin and Connor, 2017), ketogenesis (Lane et al., 2002), and energy metabolism (Goosen, 1976). In calves exposed to high maternal BHB, key ketogenesis-related genes (*HMGCS2*, *SLC16A1*) were downregulated (Figures 3B and 5B), indicating disruption of epithelial energy metabolism in the rumen. These findings align with previous reports in lambs from NEB ewes, which showed shorter rumen papillae and decreased expression of genes related to rumen epithelial integrity (*BDHI*, *CLDN1*, *MCT1*, *ZO-1*; Wang et al., 2024c). Collectively, our data suggest that high maternal BHB during the dry period may be linked to impair calves' rumen structural development and metabolic function.

Rumen inflammation may underlie these impairments. In our study, calves from cows with high BHB during the dry period were linked to increased expression of *NLRP3* and *NFKB1*, consistent with previous evidence that BHB is associated with activating the NLRP3 inflammasome and NF- $\kappa$ B signaling (Li et al., 2022). Additionally, BHB has been shown to be positively associated with increased expression of pro-inflammatory cytokines *TNF*, *IL6*, and *IL1B* in bovine hepatocytes under conditions of metabolic stress (Li et al., 2022). Similarly, we observed upregulation of these cytokines in calf rumen tissue from dams with high BHB during the dry period, including inflammatory factors (*NFKB1*, *NLRP3*) and pro-inflammatory cytokines (*IL1B*, *IL6*, *TNF*). The persistent increase in *MUC1* expression may reflect chronic mucosal stress induced by inflammatory signaling (Choudry et al., 2012). Increased mucin levels in the current study (Fang et al., 2021) suggest that the rumen of calves from the maternal high-BHB group is inflamed. Collectively, these results suggest that high maternal BHB is linked to disrupted calf rumen development through inflammation-driven pathways.

Combining inflammation and lipid metabolism signaling, PPAR signaling appears to play a critical role in postnatal rumen development (Pokhrel et al., 2025). We found downregulation of *PPAR*-regulated genes, particularly *PPAR $\gamma$* , in MB calves. To date, 3 isoforms of PPAR have been identified: *PPAR $\alpha$* , *PPAR $\beta/\delta$* , and *PPAR $\gamma$*  (Janani and Ranjitha Kumari, 2015; Mirza et al., 2019). *PPAR $\gamma$*  is highly expressed in adipose tissue and rumen, regulates lipid metabolism and inflammatory responses (Vázquez-Carrera and Wahli, 2022), and is considered a potential drug target for metabolic disorders in humans (Shi et al., 2020). In our study, inhibition of *PPAR $\gamma$*  in MB calves may exacerbate inflammation by removing its suppressive effects on NF- $\kappa$ B, thereby impairing lipid utilization and epithelial development in calves. In agreement with prior publications, *PPAR $\gamma$*  inhibition exacerbates inflammation by increasing the suppression of NF- $\kappa$ B (Blanquart et al., 2003). These results suggest that maternal circulating BHB during the dry period may disrupt rumen development of her offspring partly through *PPAR $\gamma$*  and NF- $\kappa$ B cross-talk. Overall, results of this study support the hypothesis that only slightly elevated dam BHB and NEB during the dry period may have detrimental effects on early life rumen development of her calves.

Maternal metabolic status is an important yet understudied factor in developmental programming, and the authors effectively highlight this gap. The experimental design is robust, featuring well-defined maternal BHB groups, repeated metabolic measurements, and balanced sampling of neonatal and 1-mo-old calves. The integration of rumen morphology, gene expression, and



protein-level validation enhances the reliability of the findings. This study excels in mechanistic interpretation, particularly in describing associations between higher maternal BHB levels and changes in gene and protein markers typically involved in PPAR $\gamma$ -mediated metabolic regulation and inflammatory responses. The consistency among molecular, histological, and biochemical results strengthens confidence in the conclusion that maternal metabolic stress influences postnatal rumen function.

Despite its strengths, the primary limitation of the current study is that the study relies on associative rather than causative evidence. Maternal BHB levels were naturally occurring, not experimentally manipulated, leaving potential confounding factors unresolved. The manuscript primarily reports rumen barrier-related markers, providing only indirect evidence of altered epithelial integrity. Although sample sizes are statistically justified, they may still limit generalizability, especially given biological variability in rumen development. Finally, the long-term physiological consequences require further direct measurement.

## CONCLUSIONS

This study demonstrates that high maternal BHB concentrations during the dry period, even less than the subclinical range are associated with reduced calf rumen development and function without altering birth weight. Calves from higher-BHB dams showed reduced rumen expression of genes related to epithelial integrity, differentiation, and key metabolic markers, along with increased inflammatory mediators. Collectively, the data indicate that even modest maternal BHB elevations may program rumen epithelial dysfunction through PPAR $\gamma$ -NF- $\kappa$ B cross-talk. These findings highlight the rumen's sensitivity to maternal metabolic status and provide mechanistic insight into how maternal BHB levels may influence postnatal rumen development. Importantly, they also suggest potential nutritional or therapeutic strategies to optimize maternal energy metabolism during the dry period to improve offspring rumen health.

## NOTES

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**Nonstandard abbreviations used:** 1 $\times$  TBST = Tris-buffered saline containing 0.1% Tween 20; CON = low serum BHB group; H&E = hematoxylin and eosin; HRP = horseradish peroxidase; MB = high serum BHB group; NEB = negative energy balance; PFA = paraformaldehyde; qPCR = quantitative PCR; Rel. = relative; SB = stratum basale; SC = stratum corneum; SG = stratum granulosum; SS = stratum spinosum.

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